Cross-Lingual TTS for Indian Languages: A Benchmark and Inference-Time Duration Estimation Strategies

Adarsh Arigala* · **Arjun Gangwar*** · **Srinivasan Umesh**
SPRING Lab Indian Institute of Technology, India

1. Overview

We investigate cross-lingual Text-to-Speech (TTS) in two directions: **English reference → Indic speech generation** and **Indic reference → English speech generation**. We introduce a standardized evaluation benchmark covering English and nine Indian languages and propose language-aware inference-time duration estimation strategies that incorporate language-specific duration-per-character statistics to improve speech synthesis without retraining or architectural modifications.

2. Motivation

Duration Mismatch
Alignment-free TTS models often struggle in cross-lingual scenarios due to differences in character durations across languages.

UTF-8 Based Estimation
F5-TTS sizes the output from UTF-8 byte counts, which misfires across writing systems.

Need for a Simple Solution
Existing methods require explicit duration predictors, increasing model complexity.

3. Contributions

1. **Cross-lingual benchmark**
English and nine Indian languages, evaluated in both directions.

2. **Language-aware duration modelling**
Heuristic-1 and Heuristic-2 use duration-per-character statistics

3. **No retraining required**
Works on alignment-free TTS with no architectural change.

4. **Improved performance**
Better naturalness, intelligibility and speaker similarity.

4. Methodology

4.1 Original F5-TTS

Estimates the target duration from the number of UTF-8 bytes obtained by encoding the input text a unit that is not comparable across writing systems.

$$D_{F5} = D_{ref} + \left[\frac{D_{ref}}{L_{ref}} \cdot \frac{L_{gen}}{s}\right]$$

4.2 Heuristic-1: Language-Aware Duration Model

$$D_{H1} = D_{ref} + \alpha\left[\frac{D_{ref}}{L_{ref}} \cdot \frac{L_{gen}}{s} + \beta(D_{ind} + D_{eng})\right]$$

where α = 0.7 and β = 0.3 in all experiments, D_{ind} & D_{eng} are duration estimates for Indian languages and English

4.3 Heuristic-2: Additive Duration-per-Character Guided Scaling

$$D_{H2} = D_{ref} + \left(\frac{D_{ref}}{C_{ref}} + \frac{d^c_{gen}}{d^c_{ref}}\right)\frac{C_{gen}}{s}$$

Applies relative duration-per-character differences between the reference and target languages through an additive correction.

4.4 Language-Specific Statistics

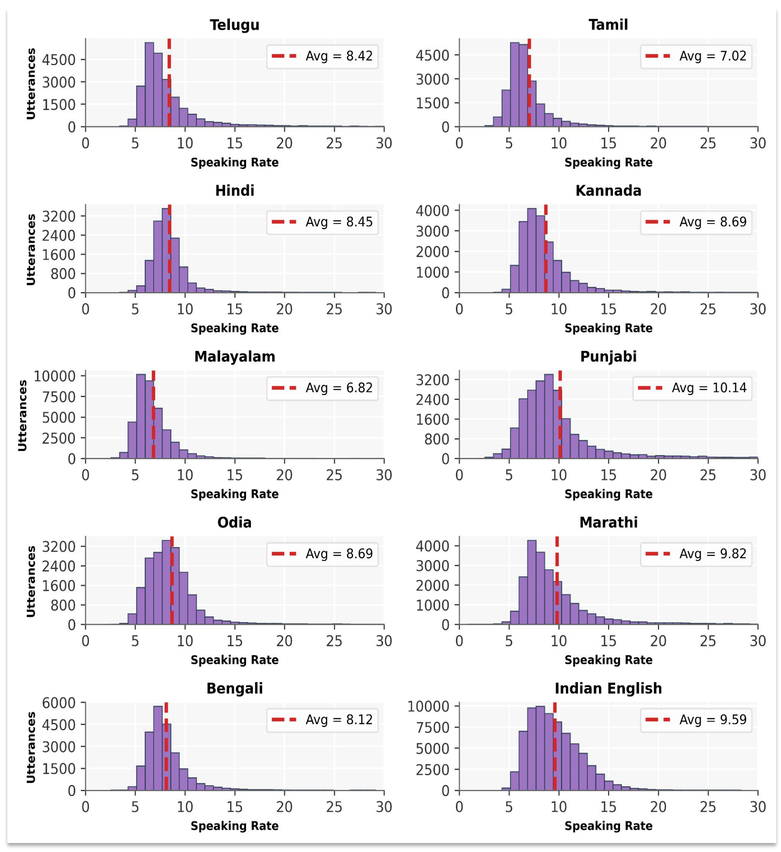


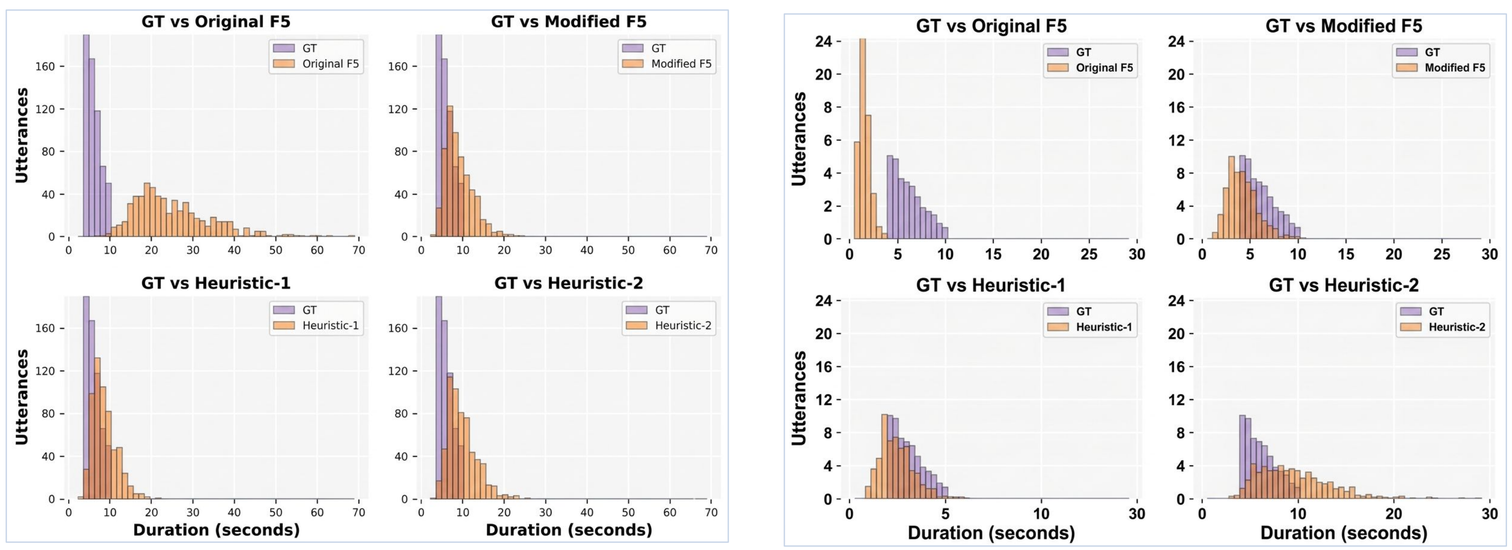
Figure 1: Utterance-level duration-per character distributions for the multilingual training data. The red dashed line indicates the average duration-per character used in the proposed heuristics.

4.5 Cross-Lingual TTS Evaluation

Model	English ref → Indic gen		Indic ref → English gen	
	WER ↓	CER ↓	WER ↓	CER ↓
Kannada				
Original F5-TTS	55.88	22.22	49.66	44.33
Modified F5	26.45	4.17	3.36	1.06
Heuristic-1	26.22	4.16	**3.18**	**0.99**
Heuristic-2	**25.15**	**3.91**	3.26	1.01
Telugu				
Original F5-TTS	37.66	18.02	45.64	41.46
Modified F5	16.73	2.08	3.65	1.20
Heuristic-1	16.40	**1.93**	3.73	**1.18**
Heuristic-2	**16.16**	1.96	**3.50**	1.25
Hindi				
Original F5-TTS	26.70	17.05	41.52	35.34
Modified F5S	11.80	3.61	4.97	1.93
Heuristic-1	**11.66**	**3.42**	4.91	2.00
Heuristic-2	11.71	3.49	**4.70**	**1.89**

Table 3 · Cross-lingual TTS evaluation, multilingual A-DMA model. Best value per column within each language in green.

5. Results



English reference → Tamil generation | Tamil reference → English generation

Figures 2 & 3 · Ground-truth (GT) and predicted duration distributions. Original F5-TTS under-estimates duration badly; character-level counting narrows the gap; both heuristics align the predicted distribution with the ground truth.

6. Key Findings

Significant reductions

- **English Reference → Indic Generation:** On Kannada, Modified F5-TTS reduces WER/CER from **52.7%/81.2%** in the multilingual setting and **75.37%/ 91.47%** in the bilingual setting.
- **Indic Reference → English Generation:** On Kannada, Modified F5-TTS achieves relative WER/CER reductions of **93.2%/97.6%** (multilingual) and **91.6%/96.1%** (bilingual).

Heuristic-1 & Heuristic-2

- **English Reference → Indic Generation:** Our proposed heuristics further reduce WER/CER by **0.9%/0.2% (bilingual)** and **4.9%/6.2% (multilingual)** compared to Modified F5-TTS.
- **Indic Reference → English Generation:** Our proposed heuristics achieve additional WER/CER reductions of **5.4%/6.6% (bilingual)** and **3.0%/4.7% (multilingual)** over Modified F5-TTS.

Naturalness preserved

Speaker similarity stays comparable while UTMOS improves.

Scan to Access the Evaluation Dataset

Scan to Explore Our TTS Demo

Scan to Visit SPRING Lab, IIT Madras